

AI AUTOMATIONS

Shayan Core Project Guide

Beginner-friendly documentation | What | Why | How | Lessons

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1. START HERE

Software = instructions for a computer.

Application = software made to perform a job.

Website = pages/files loaded by a browser.

Our system connects a public portfolio, live dashboard, iOS app, widget, GitHub and CI.

Mental model: website = information; dashboard = monitoring; app = control center; GitHub = memory; CI = automatic checker.

2. WEBSITE + DASHBOARD

HTML = HyperText Markup Language: page structure.

CSS = Cascading Style Sheets: appearance and layout.

JavaScript: behavior and interaction.

GitHub Pages publishes our website files.

Dashboard = a screen that gathers important information.

HTTP = HyperText Transfer Protocol. Our app sends a GET request to the public portfolio.

HTTP 200 = successful response. Response time is measured in ms; 1,000 ms = 1 second.

We changed the health check from the admin page to the public portfolio root.

3. GIT + CI/CD

Git = version-control system that remembers changes.

GitHub = online platform hosting our repository.

Repository = project storage. Commit = saved checkpoint. Branch = separate development line.

CI = Continuous Integration: automatically check code.

CD = Continuous Delivery/Deployment: automate preparing or releasing software.

GitHub Actions runs our workflow on macOS with Xcode.

Flow: checkout -> validate -> build -> package IPA -> sanity check -> artifact.

4. iOS APP + ARCHITECTURE

Xcode = Apple development environment. Swift = Apple programming language. SwiftUI = Apple UI framework.

View = piece of UI. NavigationStack = screen navigation.

Home is now a compact launcher; Dashboard and Website cards open full detail screens.

Core Command is the central search/action area. Modules hold larger capabilities.

The app uses live network checks and subtle pulse animations for status feedback.

5. QR SCANNER + LEARNING HUB

QR = Quick Response code. AVFoundation handles camera capture and QR metadata detection.

NSCameraUsageDescription explains why iOS needs camera access.

After scanning: show result, copy, share, or open HTTPS.

Learning Hub is now live and opens AI Automations.pdf from the app bundle.

The PDF contents page explains the project from zero and documents what, why, how and lessons.

6. TROUBLESHOOTING + HISTORY

Read the first meaningful error. Identify the layer: code, plist, project, asset, workflow or installation.

Make the smallest fix. Re-check the complete flow. Document the lesson.

Project rule: Build -> Check -> Learn -> Document -> Improve.

Current direction: portfolio + live dashboard + iOS command center + widget + QR scanner + Learning Hub.

Future meaningful features should be documented in this guide.